Department of Information Technology

Academic Year 2025 – 2026 (Even Semester)

Year,Degree,& Branch : VI-Semester-,B.Tech-IT

Course Code & Title : CCS374-&Web Application Security

Name of the Faculty member : B.Thevahi,AP/IT

Innovative Practice Description

- **Unit / Topic : I/ Security Socket Layer**
- **Course Outcome :** CO1
- **Activity Chosen : Mind Map**
- **Justification:**

  Mind maps help learners organize complex topics visually, making it easier to understand relationships between concepts and subtopics. This activity encourages critical thinking, brainstorming, and student participation, rather than passive memorization.

- **Time Allotted for the Activity:** 15 Minutes
  - ❖ **Details of the Implementation:**
  - ➢ The Faculty, taken class about Secure Socket Layer and discussed about the protocols and functions are present there.
  - ➢ Then the faculty instructed the students to draw mind map what they have understood by the class taken.
  - ➢ The students discussed with neighboring students and drew diagrams related to the Session Socket Layer.
  - ➢ That the mind map reflected the capability of students what they have understood about the protocol functions properly.

- **CO – PO / PSO mapping:**

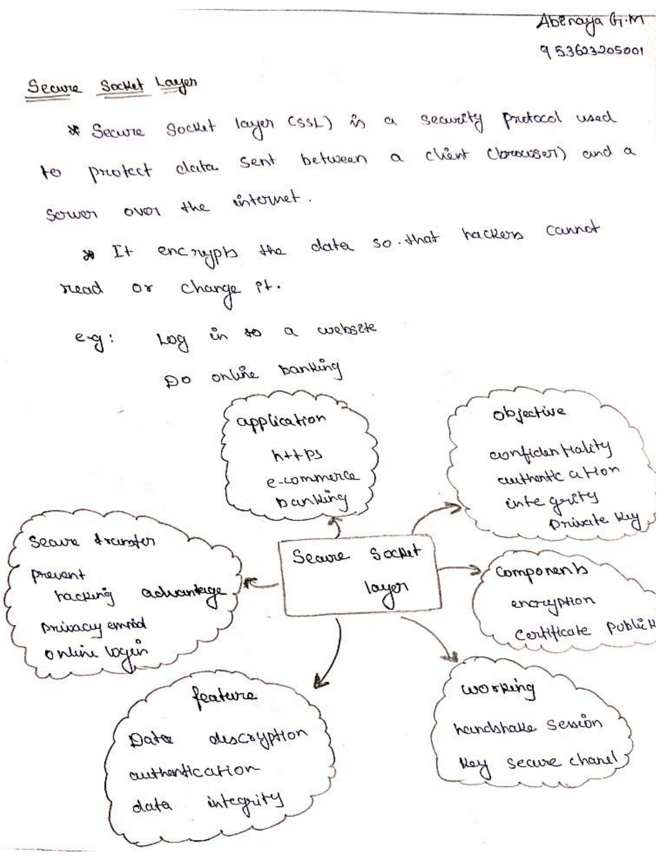
CO	PO1	PO2	PO4	PO6	PO10	PO12	PSO2	PSO3
CO1	1	2	1	2	2	2	1	1

(1 – Low 2 – Moderate 3 – High)

PO / PSO mapped:

Innovative practice	PO1	PO2	PO4	PO6	PO10	PO12	PSO2	PSO3
Justification for correlation	Students will be able to understand and apply basic concepts of network security**,** encryption, and authentication used in SSL for secure data communication.	Students will be able to analyze security problems in data transmission and identify how SSL **solves** issues such as data theft, tampering, and unauthorized access	By creating mind maps, students explore SSL components like certificates, keys, and handshake process, helping them understand how secure communication works in real systems	Students will understand the importance of SSL in protecting user privacy, online transactions, and sensitive information, ensuring safe and ethical use of internet services	Students will be able to clearly explain SSL concepts using diagrams and mind maps, improving their ability to communicate technical ideas effectively.	The activity motivates students to stay updated with modern security technologies and encourages self-learning about evolving web security standards.	Students gain knowledge of security technologies used in web, IoT, and mobile applications**,** helping them design secure and reliable systems.	Students understand the role of SSL in building secure, scalable, and trustworthy software systems, aligning with industry standards and best practices.

- **Images / Screenshot of the practice:**



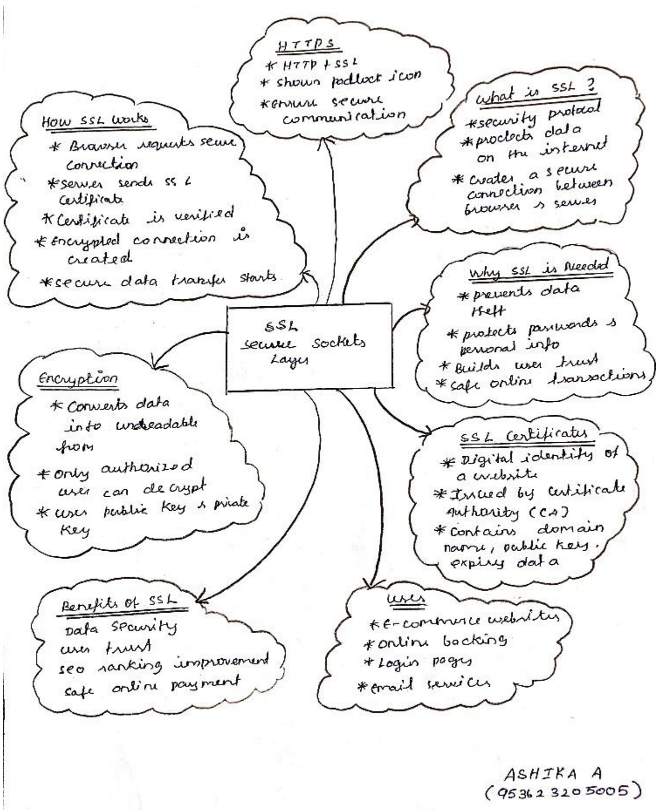


Figure1: Abinaya G.M (Sample) **Figure2:Mindmap-Ashika.A**

- **Reflective Critique:**

  ❖ ***Feedback of practice from students and other stakeholders:***

  - Students said the mind map helped them understand SSL concepts easily.
  - They felt the visual method made learning more interesting.
  - Students liked connecting theory with real-life secure websites.
  - Faculty members appreciated the innovative and interactive teaching method.
  - The activity improved student participation and learning outcomes.

  ❖ ***Benefit of the practice:***

  - The mind map activity helps students understand Secure Socket Layer (SSL) concepts in a clear and organized way
  - Visual representation improves memory and quick recall of important security components like encryption and certificates
  - The practice increases student involvement and interest during the learning process.

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  ❖ ***Challenges faced in implementation:***

  - Some students initially found it difficult to understand **basic** security and encryption concepts related to SSL.
  - A few students needed extra guidance to create clear and meaningful mind maps**.**
  - Limited class time made it challenging to cover both theory and activity in one session.
  - Differences in student learning levels required additional explanation and support**.**
  - Assessing each student's understanding during the activity required extra effort from the faculty.

References:

- https://www.ritrjpm.ac.in/images/B.Tech-IT/2024-2025/SS_IT3401_MindMap.pdf
- https://www.ritrjpm.ac.in/images/B.Tech-IT/2024-2025/KP_CCS343_Mindmap.pdf

Signature of Faculty Member **HOD**